
Dataset Distillation by Matching Training Trajectories

[Paper](#): **In Proc.** CVPR2022, George
Cazenavette et al.

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Dataset Distillation

- Datasets used in Modern Deep Learning is Huge(!)
- Huge dataset increases training cost, e.g., taking longer time.
- There are privacy issues in huge dataset.
- Large storage is required.



IG-1B

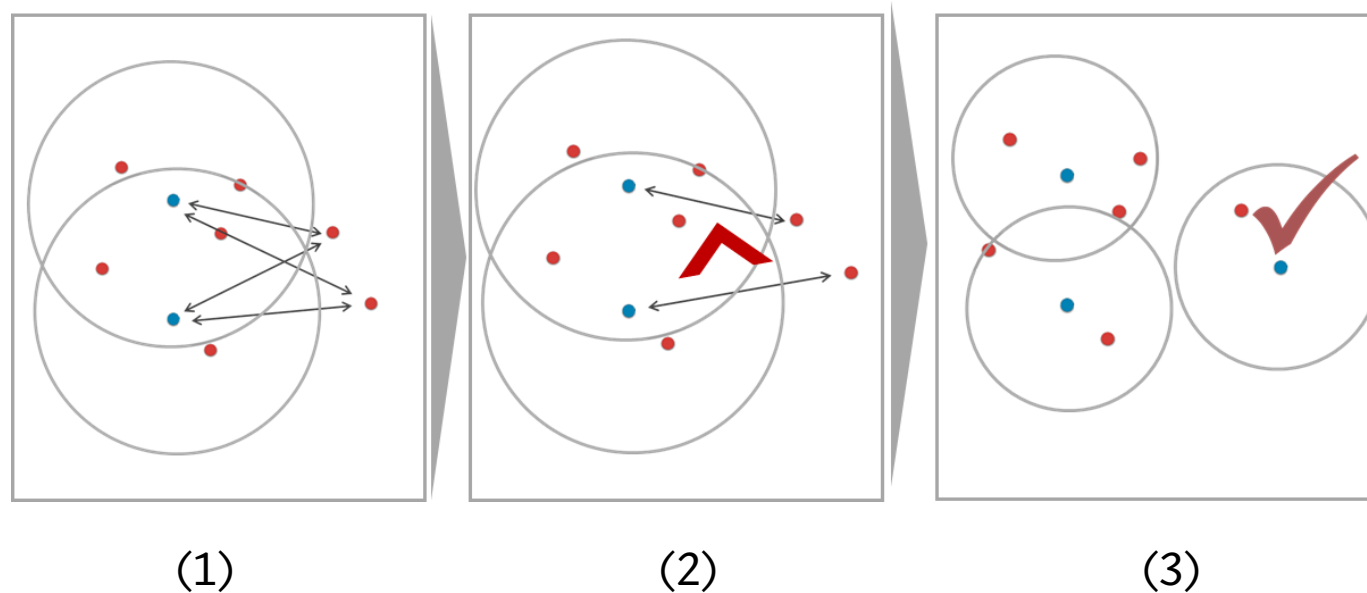


CIFAR10

Can we approximate Huge dataset by small dataset?



One Example: Coreset



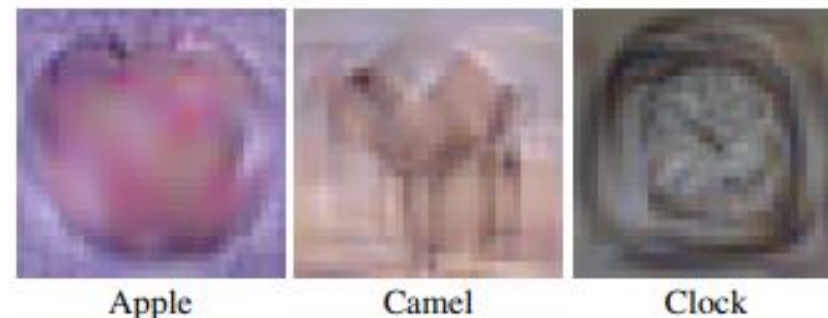
CORESET (K-Center Greedy)

	Img/Cls	Ratio %	Coreset Selection		
			Random	Herding	Forgetting
CIFAR-10	1	0.02	14.4 ± 2.0	21.5 ± 1.2	13.5 ± 1.2
	10	0.2	26.0 ± 1.2	31.6 ± 0.7	23.3 ± 1.0
	50	1	43.4 ± 1.0	40.4 ± 0.6	23.3 ± 1.1
CIFAR-100	1	0.2	4.2 ± 0.3	8.4 ± 0.3	4.5 ± 0.2
	10	2	14.6 ± 0.5	17.3 ± 0.3	15.1 ± 0.3
	50	10	30.0 ± 0.4	33.7 ± 0.5	30.5 ± 0.3
Tiny ImageNet	1	0.2	1.4 ± 0.1	2.8 ± 0.2	1.6 ± 0.1
	10	2	5.0 ± 0.2	6.3 ± 0.2	5.1 ± 0.2
	50	10	15.0 ± 0.4	16.7 ± 0.3	15.0 ± 0.3

- Coreset prone to relying on Heuristic Method. (since the original problem is NP-hard)
- It confines itself to the samples existing in the original dataset.

Dataset Distillation

- DD synthesizes a small dataset that approximates the large dataset from scratch!



- **For now, the classification task is presumed !**

$$\mathcal{L}(\mathcal{S}, \mathcal{T}) = \mathbb{E}_{\theta^{(0)} \sim \Theta} [l(\mathcal{T}; \theta^{(T)})],$$
$$\theta^{(t)} = \theta^{(t-1)} - \eta \nabla l(\mathcal{S}; \theta^{(t-1)}),$$

The objective in the original [paper](#).



$\mathcal{T}$ is real dataset, $\mathcal{S}$ is synthetic dataset

Goal: The model trained using synthetic dataset should have high performance when tested in real training data.

Previous works

Goal: To generate **synthetic dataset** that **approximate** the original dataset

(1) Performance Matching **How can we solve this?** (2) Gradient Matching

$$\mathcal{L}(\mathcal{S}, \mathcal{T}) = \mathbb{E}_{\theta^{(0)} \sim \Theta} [l(\mathcal{T}; \theta^{(T)})],$$
$$\theta^{(t)} = \theta^{(t-1)} - \eta \nabla l(\mathcal{S}; \theta^{(t-1)}),$$

Shortcomings: Heavy computation !

$$\tilde{Y}_{\mathcal{S}}^* = \arg \min_{\tilde{Y}_{\mathcal{S}}} \sum_{\mathbf{x}, \mathbf{y} \sim \mathcal{T}} L(f_{\Theta'}(\mathbf{x}), \mathbf{y})$$

Similar: Label distillation

$$\mathcal{L}(\mathcal{S}, \mathcal{T}) = \mathbb{E}_{\theta^{(0)} \sim \Theta} \left[\sum_{t=0}^T \mathcal{D}(\mathcal{S}, \mathcal{T}; \theta^{(t)}) \right],$$
$$\theta^{(t)} = \theta^{(t-1)} - \eta \nabla l(\mathcal{S}; \theta^{(t-1)}),$$

$$\mathcal{D}(\mathcal{S}, \mathcal{T}; \theta) = \sum_{c=0}^{C-1} d(\nabla l(\mathcal{S}_c; \theta), \nabla l(\mathcal{T}_c; \theta)),$$

$$d(\mathbf{A}, \mathbf{B}) = \sum_{i=1}^L \sum_{j=1}^{J_i} \left(1 - \frac{\mathbf{A}_j^{(i)} \cdot \mathbf{B}_j^{(i)}}{\|\mathbf{A}_j^{(i)}\| \|\mathbf{B}_j^{(i)}\|} \right),$$



Previous works (2)

(3) Distribution Matching

$$\mathcal{L}(\mathcal{S}, \mathcal{T}) = \mathbb{E}_{\theta \in \Theta} [\mathcal{D}(\mathcal{S}, \mathcal{T}; \theta)]$$

$$\mathcal{D}(\mathcal{S}, \mathcal{T}; \theta) = \sum_{c=0}^{C-1} \|\mu_{\theta,s,c} - \mu_{\theta,t,c}\|^2,$$

$$\mu_{\theta,s,c} = \frac{1}{M_c} \sum_{j=1}^{M_c} f_{\theta}^{(i)}(X_{s,c}^{(j)}), \quad \mu_{\theta,t,c} = \frac{1}{N_c} \sum_{j=1}^{N_c} f_{\theta}^{(i)}(X_{t,c}^{(j)}),$$

- **Avoid** expensive computation stemming from bi-level optimization
- Performance **degrades** from SOTA

To summarize..

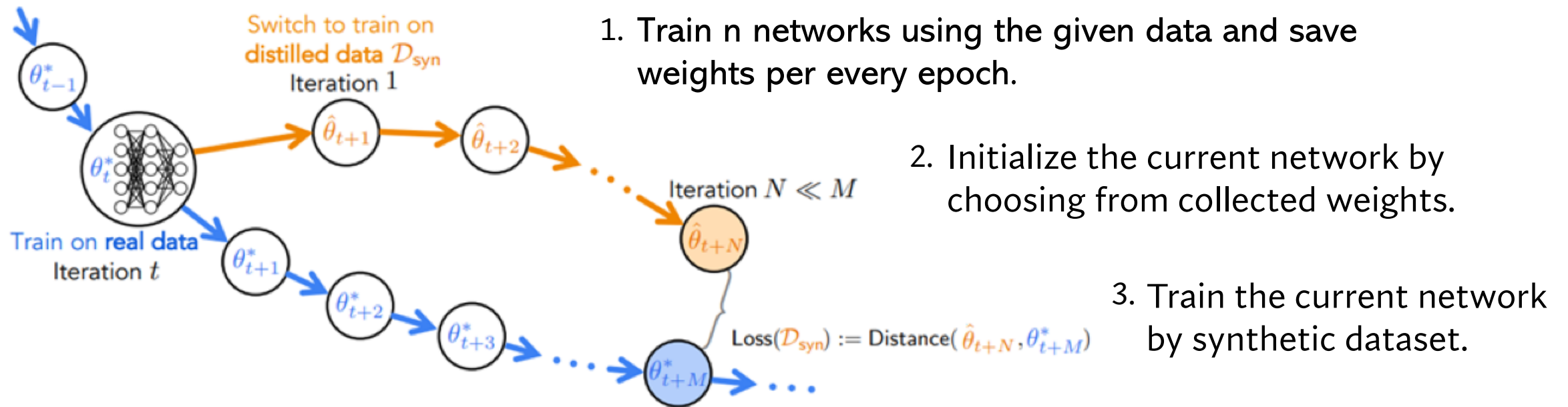
- Performance matching is **short-sighted** and is **difficult** to optimize
- Gradient matching has the **similar** issue.
- While distribution matching alleviate this, its performance **degrades**.

Short-sighted..



Method

- **Idea:** Match the weights in real data training and synthetic data training!



4. **Match** the weights in step 3 to the succeeding weights in real data training.

Method(In detail)

- Collect $\tau_1 \dots \tau_n$ real data training trajectories: $\tau = \{\theta_t^*\}_0^T$
- Set current weights $\hat{\theta}_t \leftarrow \theta_t$ by sampling from the collected trajectories, where $t < T_+$
- Gradient-descend N-steps using D_{syn} to obtain $\hat{\theta}_{t+N}$

$$\hat{\theta}_{t+n+1} = \hat{\theta}_{t+n} - \alpha \nabla \ell(\mathcal{A}(\mathcal{D}_{syn}); \hat{\theta}_{t+n}),$$


Gradient flows to here!

- Obtain θ_{t+M} from the training trajectory and minimize the following:

$$\mathcal{L} = \frac{\|\hat{\theta}_{t+N} - \theta_{t+M}^*\|_2^2}{\|\theta_t^* - \theta_{t+M}^*\|_2^2},$$


L2 loss between parameters with Regularization !

```

1: Initialize distilled data  $\mathcal{D}_{\text{syn}} \sim \mathcal{D}_{\text{real}}$  (1)
2: Initialize trainable learning rate  $\alpha := \alpha_0$  for apply  $\mathcal{D}_{\text{syn}}$ 
3: for each distillation step... do
4:     ▷ Sample expert trajectory:  $\tau^* \sim \{\tau_i^*\}$  with  $\tau^* = \{\theta_t^*\}_0^T$ 
5:     ▷ Choose random start epoch,  $t \leq T^+$ 
6:     ▷ Initialize student network with expert params:
7:          $\hat{\theta}_t := \theta_t^*$ 
8:     for  $n = 0 \rightarrow N - 1$  do
9:         ▷ Sample a mini-batch of distilled images:
10:             $b_{t+n} \sim \mathcal{D}_{\text{syn}}$  (2)
11:         ▷ Update student network w.r.t. classification loss:
12:             $\hat{\theta}_{t+n+1} = \hat{\theta}_{t+n} - \alpha \nabla \ell(\mathcal{A}(b_{t+n}); \hat{\theta}_{t+n})$  (3)
13:     end for
14:     ▷ Compute loss between ending student and expert params:
15:         $\mathcal{L} = \|\hat{\theta}_{t+N} - \theta_{t+M}^*\|_2^2 / \|\theta_t^* - \theta_{t+M}^*\|_2^2$ 
16:     ▷ Update  $\mathcal{D}_{\text{syn}}$  and  $\alpha$  with respect to  $\mathcal{L}$  (4)
17: end for

```

- 1) Initialize synthetic dataset using the portion of real data
- 2) Sampling from synthetic dataset during N-step GD.
✓ This reduces the memory consumption significantly
- 3) DiffAugmentation
- 4) Notice the learning rate is also learnt!

Questions?

Visual results

- Experiments were conducted in CIFAR-10/100, Tiny-ImageNet, ImageNet subsets.
- ConvNet was employed, which consists of several blocks, each containing 3×3 / 128 kernels, ReLU, 2×2 Average Pooling.



Experiments

			Training Set Synthesis								Full Dataset
	Img/Cls	Ratio %	DD [†] [44]	LD [†] [2]	DC [47]	DSA [45]	DM [46]	CAFE [43]	CAFE+DSA [43]	Ours	
CIFAR-10	1	0.02	-	25.7 ± 0.7	28.3 ± 0.5	28.8 ± 0.7	26.0 ± 0.8	30.3 ± 1.1	31.6 ± 0.8	46.3 ± 0.8*	84.8 ± 0.1
	10	0.2	36.8 ± 1.2	38.3 ± 0.4	44.9 ± 0.5	52.1 ± 0.5	48.9 ± 0.6	46.3 ± 0.6	50.9 ± 0.5	65.3 ± 0.7*	
	50	1	-	42.5 ± 0.4	53.9 ± 0.5	60.6 ± 0.5	63.0 ± 0.4	55.5 ± 0.6	62.3 ± 0.4	71.6 ± 0.2	
CIFAR-100	1	0.2	-	11.5 ± 0.4	12.8 ± 0.3	13.9 ± 0.3	11.4 ± 0.3	12.9 ± 0.3	14.0 ± 0.3	24.3 ± 0.3*	56.2 ± 0.3
	10	2	-	-	25.2 ± 0.3	32.3 ± 0.3	29.7 ± 0.3	27.8 ± 0.3	31.5 ± 0.2	40.1 ± 0.4	
	50	10	-	-	-	42.8 ± 0.4	43.6 ± 0.4	37.9 ± 0.3	42.9 ± 0.2	47.7 ± 0.2*	
Tiny ImageNet	1	0.2	-	-	-	-	3.9 ± 0.2	-	-	8.8 ± 0.3	37.6 ± 0.4
	10	2	-	-	-	-	12.9 ± 0.4	-	-	23.2 ± 0.2	
	50	10	-	-	-	-	24.1 ± 0.3	-	-	28.0 ± 0.3	

For CIFAR10/CIFAR100, **depth-3 ConvNet** was used and **ZCA-whitening** was applied.

ZCA-whitening?

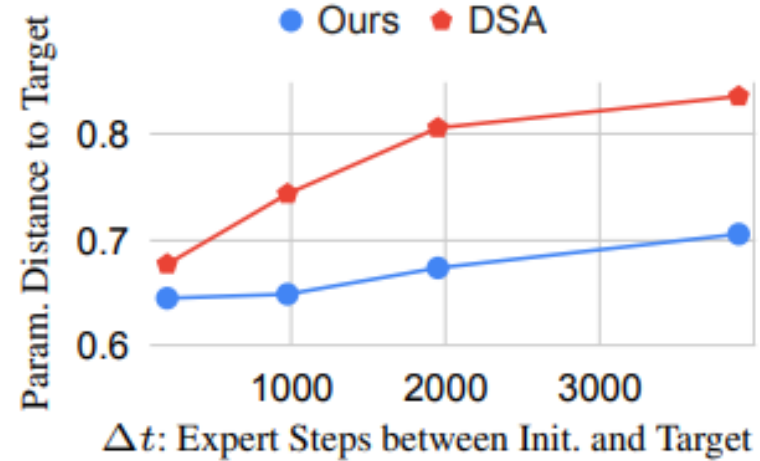
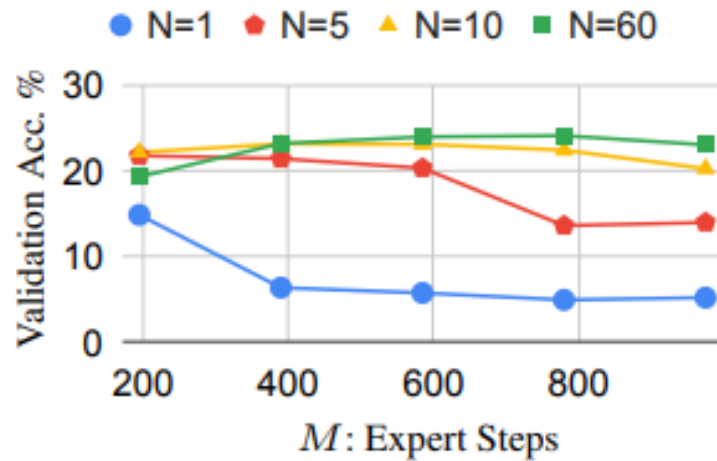


Make feature less correlated!

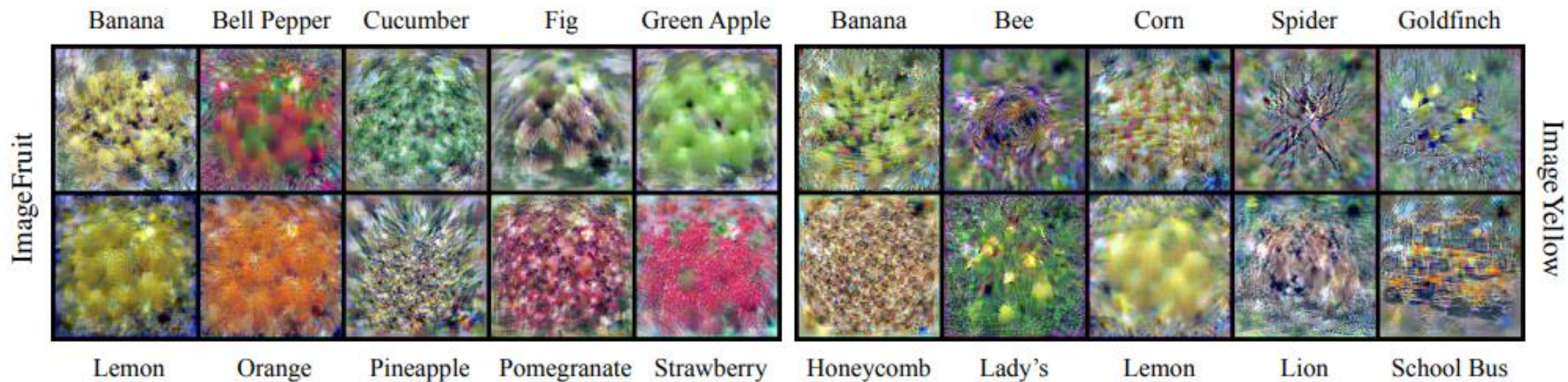


		Evaluation Model			
		ConvNet	ResNet	VGG	AlexNet
Method	Ours	64.3 ± 0.7	46.4 ± 0.6	50.3 ± 0.8	34.2 ± 2.6
	DSA	52.1 ± 0.4	42.8 ± 1.0	43.2 ± 0.5	35.9 ± 1.3
	KIP	47.6 ± 0.9	36.8 ± 1.0	42.1 ± 0.4	24.4 ± 3.9

Cross Architecture Generalization (CIFAR10 / 10 imgs)



Optimal synthetic data steps & Parameter divergence (CIFAR100 / 1 img)

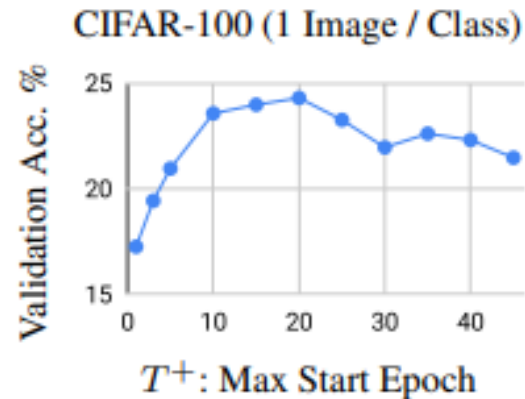
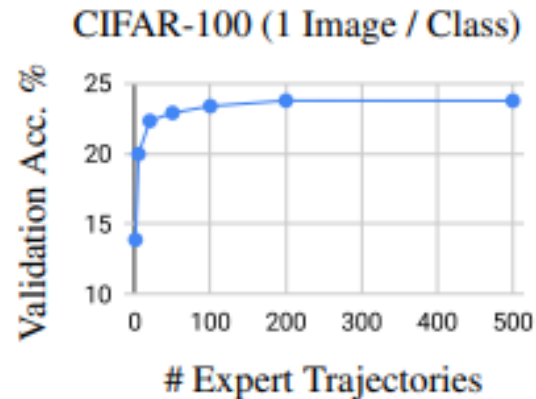


Distilled dataset is dependent to other samples! → observe the color change in Banana sample.

ImageNet subset results

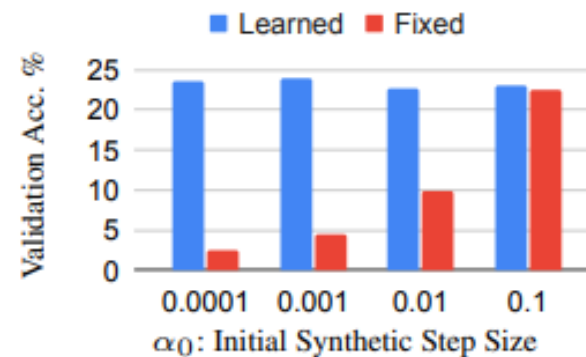
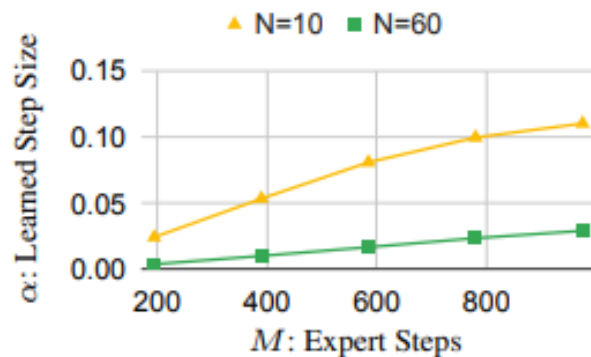
	ImageNette	ImageWoof	ImageFruit	ImageMeow	ImageSquawk	ImageYellow
1 Img/Cls	47.7 ± 0.9	28.6 ± 0.8	26.6 ± 0.8	30.7 ± 1.6	39.4 ± 1.5	45.2 ± 0.8
10 Img/Cls	63.0 ± 1.3	35.8 ± 1.8	40.3 ± 1.3	40.4 ± 2.2	52.3 ± 1.0	60.0 ± 1.5
Full Dataset	87.4 ± 1.0	67.0 ± 1.3	63.9 ± 2.0	66.7 ± 1.1	87.5 ± 0.3	84.4 ± 0.6

Ablations



Setting	Ours	Gauss. Init.	w/o ZCA	w/o Diff. Aug.
Acc.	24.3 ± 0.3	22.9 ± 0.4	19.1 ± 0.6	20.7 ± 0.7

Experiments on synthetic sample initialization



Experiments on Learning rate!

Dataset	Model	Img/Cls	Synthetic Steps (N)	Expert Epochs ($M^\dagger$)	Max Start Epoch (T^+)	Synthetic Batch Size ($ b $)	Learning Rate (Pixels)	Learning Rate (Step Size)	Starting Synthetic Step Size (α_0)
CIFAR-10	ConvNetD3	1	50	2	5	-	10^3	10^{-7}	10^{-2}
		10	30	2	15	-	10^3	10^{-5}	10^{-2}
		50	30	2	40	-	10^3	10^{-5}	10^{-3}
CIFAR-10 (ZCA)	ConvNetD3	1	50	2	2	-	10^2	10^{-7}	10^{-2}
		10	30	2	20	-	10^5	10^{-4}	10^{-2}
		50	30	2	20	-	10^2	10^{-5}	10^{-3}
CIFAR-100	ConvNetD3	1	40	3	15	-	10^3	10^{-5}	10^{-2}
		10	20	2	40	-	10^3	10^{-5}	10^{-2}
		50	80	2	40	125	10^3	10^{-5}	10^{-2}
CIFAR-100 (ZCA)	ConvNetD3	1	20	3	20	-	10^3	10^{-5}	10^{-2}
		10	20	2	20	-	10^3	10^{-5}	10^{-2}
		50	80	2	40	125	10^3	10^{-5}	10^{-2}
Tiny ImageNet	ConvNetD4	1	10	2	10	-	10^4	10^{-4}	10^{-2}
		10	20	2	40	200	10^4	10^{-4}	10^{-2}
		50	20	2	40	300	10^4	10^{-4}	10^{-2}
ImageNet (All)	ConvNetD5	1	20	2	10	-	10^4	10^{-6}	10^{-2}
		10	20	2	10	20	10^5	10^{-6}	10^{-2}

Evaluation Epoch of Distilled data: 1000

Questions?
